

To what extent can assistive AI tools improve equitable digital accessibility and inclusive participation for marginalized communities, specifically individuals with disabilities and those from low-income backgrounds?

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**Abstract:**

Access to technology is not equally distributed, disproportionately impacting marginalized groups with limited resources or abilities. The digital divide remains a persistent global issue, as many communities lack access to reliable internet, affordable devices, or user-friendly software. This problem is particularly present for people with disabilities and those from low-income or underrepresented backgrounds, significantly limiting their digital accessibility. Artificial intelligence (AI) presents new opportunities to make communication, participation, and digital access equitable. This study aims to investigate the extent to which AI tools can effectively help mitigate the digital divide. To achieve this, 60 participants from marginalized communities identified through disability support organisations, community groups, and income-based eligibility assessments will be divided into two groups. Group A will utilise AI-enhanced tools such as screen readers (e.g. NVDA, VoiceOver) and speech-to-text software (e.g. Google Voice Typing, Windows Dictation), while Group B will use standard tools. Both groups will complete identical digital tasks. We expect the results to demonstrate that AI tools significantly improve accessibility and inclusion for marginalized groups compared to standard tools by reducing barriers related to literacy, visual impairments, and motor disabilities, resulting in greater digital independence. These findings will highlight the potential and importance of artificial intelligence in reducing digital inequalities and promoting greater accessibility.

**Keywords:**

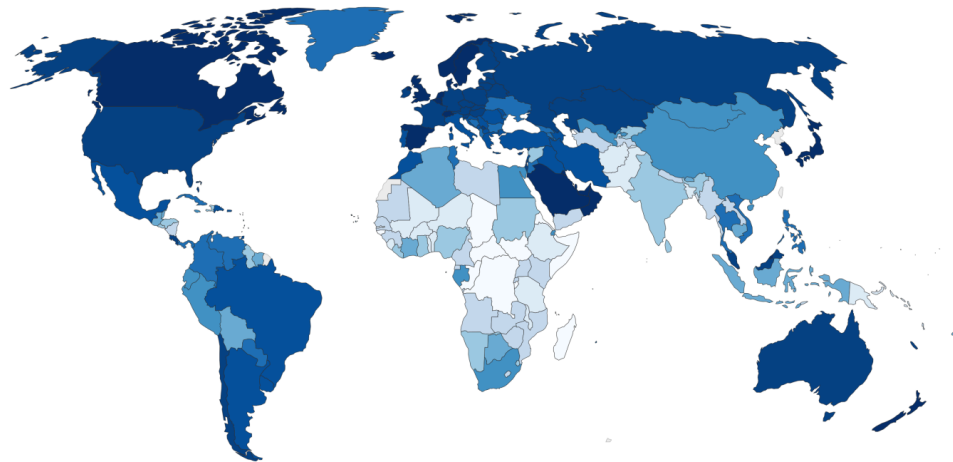
Digital accessibility, assistive technologies, marginalized groups, human-computer interaction

**Scientific Background:**

The digital divide refers to the unequal access to modern digital technologies, including computers and reliable internet connectivity. This inequality, shaped by socioeconomic and infrastructural factors, disproportionately affects marginalized groups, including people with disabilities and low-income individuals. For instance, according to Eurostat (2025), internet access among individuals with disabilities in the EU is 15% lower than those without disabilities, particularly impacting the elderly.

### Share of the population using the internet, 2019

All individuals who have used the Internet in the last 3 months are counted as Internet users. The Internet can be used via a computer, mobile phone, personal digital assistant, gaming device, digital TV etc.



Source: International Telecommunication Union (via World Bank)

[OurWorldInData.org/technology-adoption/](https://OurWorldInData.org/technology-adoption/) • CC BY

**Figure 1:** The disparity in access to the internet (ITU)

Globally, about 2.6 billion people still lack access to the internet, despite significant advancements in digital infrastructure. Only 27% in low-income countries have access, compared to 93% in high-income ones. In some areas, just 1 in 100 people have reliable connectivity, while 73% of low-income households remain disconnected. Moreover, over 1

billion people need assistive technologies — such as screen readers or speech-to-text software, to access digital content — but only 20% have access to such tools.

These inequalities highlight the urgent need for inclusive solutions. Artificial intelligence (AI) and other emerging technologies offer various opportunities to address these barriers and promote greater digital equity.

**Hypothesis:**

*If* assistive artificial intelligence tools are designed with accessibility features and input from marginalized users, *then* individuals with disabilities and from low-income backgrounds will communicate and participate more effectively than when using standard tools.

**Rationale:**

Artificial intelligence (AI) has the potential to address and remove barriers faced by marginalized populations, particularly individuals with disabilities and those from low-income backgrounds. For individuals with physical or sensory disabilities, these technologies can facilitate more independent navigation of online platforms and enhance communication. Similarly, individuals with low-income backgrounds often lack reliable access to educational resources and support networks. AI can simplify processes and provide affordable access to services and education, helping reduce long-term exclusion. This project focuses on these groups, recognising the challenges they face and how they overlap. Though promising, many AI-based solutions are developed without sufficient input from these groups. This experiment aims to explore whether

inclusive AI technology can provide a measurable improvement in accessibility over conventional, non-AI tools.

### **Methodology:**

Goal: To evaluate whether specific AI tools enhance communication and digital participation among individuals from marginalized communities, compared to standard non-AI tools.

Participants: This study will have 60 participants, comprising individuals with physical or sensory disabilities and individuals from low-income or underrepresented backgrounds. They will be selected through a voluntary sign-up process, ensuring demographic diversity and eligibility. Participants will randomly be assigned to one of two groups to reduce selection bias:

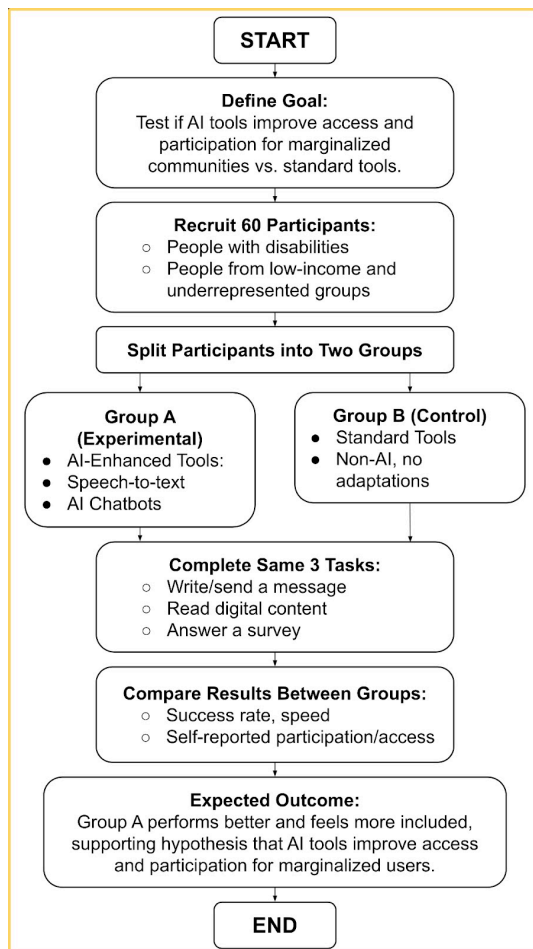
### **Design:**

Participants will be split into two equal groups (n=30):

1. **Group A** will use AI-enhanced assistive tools such as screen readers (like voiceovers, for navigating text and web content), speech-to-text software (like Google Live Transcribe, for writing or filling out forms), and adaptive AI chatbots (conversational agents like voice assistants that explain tasks or answer questions in simple language).
  - a. This group will be the experimental group, as it will test whether AI enhances communication and participation.
2. **Group B** will use standard digital tools without accessibility or adaptive AI features (e.g. regular websites, basic messaging platforms, basic PDF documents, etc)
  - a. This would be the control group, as its users utilise standard digital tools without any AI or accessibility enhancements.

Both groups' participants will complete the study over a week, with one task assigned per day across three separate sessions. For instance, these tasks would include:

1. Day 1: Write and send a message or email.
2. Day 2: Read and respond to digital content.
3. Day 3: Complete a structured survey.



Each session will be timed and observed, with standardized instructions. After each task, participants will complete a short feedback form to report on ease of use, comfort, and perceived accessibility. The independent variable in this study is the type of tool use (assistive AI vs. standard tools), while the dependent variables include task success rate, time taken, and self-reported participation and accessibility. It is expected that the experimental group using AI-enhanced tools will complete tasks more efficiently, report lower levels of frustration, and feel more included, indicating that AI tools can lead to measurable improvements in digital access and participation.

**Figure 2:** Flowchart of experimental design

### **Conclusion:**

This research shows that AI can ultimately improve the quality of life for disadvantaged populations, especially people with disabilities — like reading text out loud or translating languages. However, many still face barriers such as high costs, lack of internet, or tools that do not meet or accommodate their needs. To be truly helpful, AI must be built inclusively from the start: using diverse data, involving affected communities in the design process, testing for bias, and ensuring affordability and accessibility. With these changes, AI can reduce digital inequality and help create a more just society.

**Definition Sheet:**

1. Artificial Intelligence (AI) = The ability of computers and machines to simulate human learning and comprehension, performing tasks that typically require human intelligence. In the methodology, assistive tools used would be screen readers, speech-to-text software, and AI chatbots.
2. Disability = A physical, sensory, cognitive, or mental condition that may limit a person's ability to perform certain tasks or participate fully in daily activities. Disabilities can be visible or invisible and may be permanent, temporary, or situational. The research focuses on individuals with sensory disabilities (especially visual impairments), physical/motor disabilities (related to typing/navigation), and learning or cognitive disabilities.



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